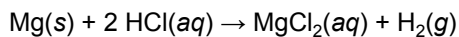


Magnesium metal reacts with hydrochloric acid according to Reaction 1.



Reaction 1

During an experiment to measure the molar enthalpy ΔH° of Reaction 1, a student placed 100. mL of dilute HCl(aq) solution into a coffee cup calorimeter, as shown in Figure 1.

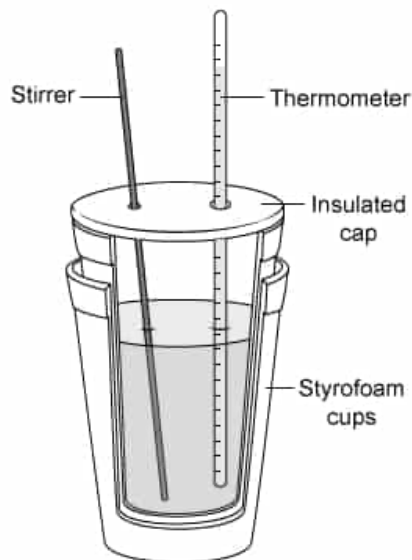


Figure 1 Coffee cup calorimeter

After monitoring the HCl(aq) solution temperature for 1 min, 0.486 g (0.0200 mol) of Mg(s) was then added to the HCl(aq) solution inside the calorimeter and stirred while continuing to monitor the temperature until the Mg(s) was completely consumed. The results of the experiment are shown on the graph in Figure 2.

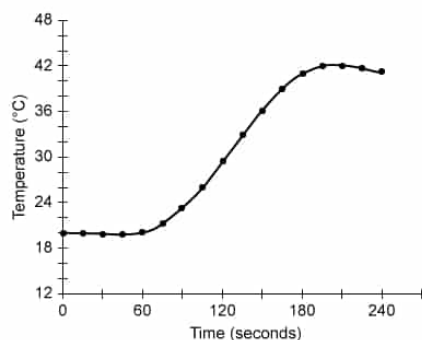


Figure 2 Graph of temperature versus time during the reaction of magnesium with hydrochloric acid

According to the first law of thermodynamics, the heat produced (or consumed) by a reaction q_{rxn} must be equal to the heat transferred to (or taken from) the surroundings (ie, the solution and the calorimeter), as expressed by Equation 1.

$$-q_{\text{rxn}} = q_{\text{surroundings}} = q_{\text{solution}} + q_{\text{calorimeter}}$$

Equation 1

Based on the results of the calorimetry experiment described in the passage, the magnitude of the molar enthalpy ΔH° for Reaction 1 (in kJ/mol Mg) is closest to which of the following? (Note: Assume that the HCl(aq) solution has a density of 1.0 g/mL and a specific heat capacity of 4.0 J/(g·°C), and that $q_{\text{calorimeter}} \approx 0$.)

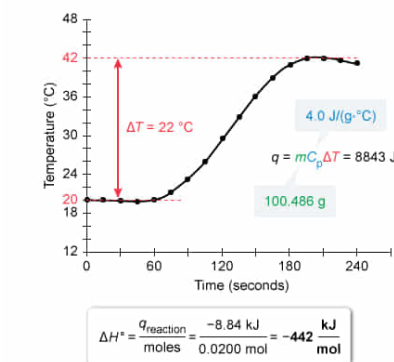
- ☐ A. 8.84 [24%]
☐ B. 18.2 [27%]
☒ C. 442 [39%]

Correct

38% answered correctly

Explanation:

Calculating molar enthalpy ΔH° using calorimetry



The **specific heat capacity** C_p of a substance (sometimes abbreviated as specific heat) is the amount of heat required to raise the temperature of 1 g of the substance by 1°C . Accordingly, if the mass m and change in temperature ΔT of a substance are known, the C_p of the substance can be used to calculate the heat q that is absorbed or released by the substance using the relationship:

$$q = mC_p(T_{\text{final}} - T_{\text{initial}}) = mC_p \Delta T$$

This relationship holds for both solids and solutions. In the experiment described in the passage, it is the change in the temperature of the HCl(aq) solution that is monitored. As such, the mass and heat capacity of the HCl(aq) solution should be used in this equation to calculate the amount of heat transferred by the reaction.

Although dissolved solutes affect both the mass and density of a solution, dilute aqueous solutions are primarily water. As such, the density of dilute solutions can often be approximated by the **density of water**, which remains very **close to 1.0 g/mL between 1°C and 100°C** . Accordingly, the mass of the 100. mL of dilute HCl(aq) solution in the calorimeter can be estimated:

$$100. \text{ mL HCl(aq)} \times \frac{1.0 \text{ g HCl(aq)}}{1 \text{ mL HCl(aq)}} = 100. \text{ g HCl(aq)}$$

Adding the mass of the Mg(s) to the mass of the HCl(aq) gives the total mass of the solution system after the magnesium is introduced. Because the amount of Mg(s) is small compared to the amount of HCl(aq) , the added mass has a negligible effect on the overall properties of the HCl(aq) solution:

$$m = 100. \text{ g} + 0.486 \text{ g} = 100.486 \text{ g}$$

Based on the temperature change data in Figure 2 and the specific heat capacity of the HCl(aq) solution, the **heat gained by the solution** in the experiment is approximately:

$$q_{\text{solution}} = (100.486 \text{ g}) \left(4.0 \frac{\text{J}}{\text{g}\cdot^\circ\text{C}} \right) (42^\circ\text{C} - 20^\circ\text{C})$$

$$q_{\text{solution}} = \left(401.944 \frac{\text{J}}{^\circ\text{C}} \right) (22^\circ\text{C}) = 8,843 \text{ J} \implies 8.84 \text{ kJ}$$

Applying Equation 1 shows that the heat gained by the HCl(aq) solution must be equal to the heat released by the reaction q_{reaction} , where the negative sign indicates that heat is released.

$$-q_{\text{reaction}} = q_{\text{solution}} + q_{\text{calorimeter}}$$

$$-q_{\text{reaction}} = 8.84 \text{ kJ} + 0 \text{ kJ}$$

$$q_{\text{reaction}} = -8.84 \text{ kJ}$$

Dividing the mass of Mg(s) by its **molar mass** from the periodic table gives the moles of Mg(s) involved in the reaction stated in the passage.

$$0.486 \text{ g Mg(s)} \times \frac{1 \text{ mol}}{24.31 \text{ g Mg(s)}} = 0.0200 \text{ mol Mg(s)}$$

Therefore, the molar enthalpy ΔH° of the reaction (in kJ/mol Mg) is:

$$\Delta H^\circ = \frac{q_{\text{reaction}}}{\text{moles}} = \frac{-8.84 \text{ kJ}}{0.0200 \text{ mol}} = -442 \frac{\text{kJ}}{\text{mol}}$$

(Choice A) 8.84 kJ is the magnitude of q_{reaction} . This value must be divided by the moles of reaction to determine ΔH° .

(Choice B) A magnitude of 18.2 is obtained by dividing q_{reaction} by the mass instead of the moles of Mg(s).

(Choice D) A magnitude of 844 is obtained if the calculation is performed using the maximum temperature ($T = 42^\circ\text{C}$) instead of the change in temperature ($\Delta T = 22^\circ\text{C}$).

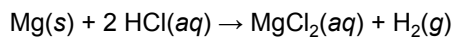
Educational objective:

Multiplying the specific heat capacity by the mass and change in temperature of a sample yields the associated amount of heat absorbed (or released) by the sample. The molar enthalpy of a reaction is calculated by dividing the heat of the reaction by the moles of the reaction.

Time spent: 0 sec
Subject:
 General Chemistry

QID: 403176
System:
 Thermodynamics, Kinetics & Gas Laws

Magnesium metal reacts with hydrochloric acid according to Reaction 1.



Reaction 1

During an experiment to measure the molar enthalpy ΔH° of Reaction 1, a student placed 100. mL of dilute HCl(aq) solution into a coffee cup calorimeter, as shown in Figure 1.

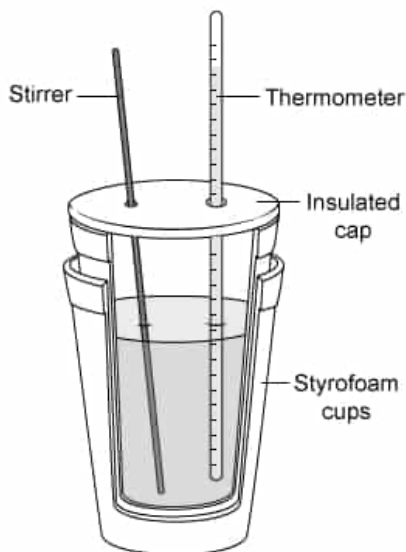


Figure 1 Coffee cup calorimeter

After monitoring the HCl(aq) solution temperature for 1 min, 0.486 g (0.0200 mol) of Mg(s) was then added to the HCl(aq) solution inside the calorimeter and stirred while continuing to monitor the temperature until the Mg(s) was completely consumed. The results of the experiment are shown on the graph in Figure 2.

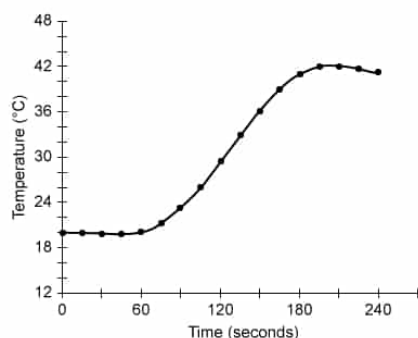


Figure 2 Graph of temperature versus time during the reaction of magnesium with hydrochloric acid

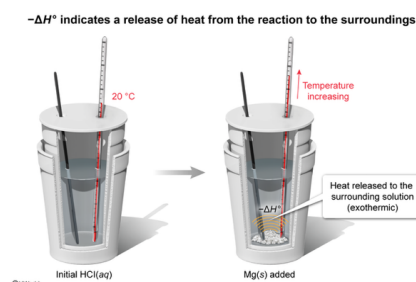
According to the first law of thermodynamics, the heat produced (or consumed) by a reaction q_{rxn} must be equal to the heat transferred to (or taken from) the surroundings (ie, the solution and the calorimeter), as expressed by Equation 1.

$$-q_{\text{rxn}} = q_{\text{surroundings}} = q_{\text{solution}} + q_{\text{calorimeter}}$$

Equation 1

Based on the results from Figure 2, the molar enthalpy of the reaction ΔH° is:

- ☐ A. positive because heat flows from the reaction system to the surroundings. [11%]
- ☐ B. positive because heat flows from the surroundings to the reaction system. [32%]
- ☒ C. negative because heat flows from the reaction system to the surroundings. [51%]
- ☐ D. negative because heat flows from the surroundings to the reaction system. [4%]

Explanation:

During a chemical reaction between two molecules, some bonds are broken as new bonds are formed. **Breaking bonds consumes energy** whereas **forming bonds releases energy**. If the amount of energy needed to break existing bonds is less than the amount released from forming new bonds, the reaction is **exothermic** and **releases the excess energy** (as heat) to the surroundings (ie, the solvent), which raises the temperature. Conversely, if more energy is consumed than released, the reaction is **endothermic** and **absorbs energy** from the surroundings, which lowers the temperature.

When measured under constant pressure, the heat released (or absorbed) by a reaction is equal to the **change in enthalpy ΔH°** of the reaction. A **negative enthalpy** indicates an exothermic reaction, and a positive enthalpy indicates an endothermic reaction.

As seen in Figure 2 in the passage, Reaction 1 causes the temperature in the calorimeter to increase, which indicates a release of thermal energy by an exothermic reaction. Therefore, ΔH° is negative for the reaction because heat flows from the reaction system to the surroundings.

(Choices A and B) A positive ΔH° indicates an endothermic reaction that absorbs energy (heat) from the surroundings, which causes the temperature to decrease. Figure 2 shows an increase in temperature.

(Choice D) The ΔH° for Reaction 1 is negative, but heat must flow from the reaction to the surroundings to produce the observed increase in temperature.

Educational objective:

A negative change in enthalpy ΔH° indicates an exothermic reaction that releases heat to the surroundings and raises the temperature. Conversely, a positive ΔH° indicates an endothermic reaction that absorbs heat from the surroundings and lowers the temperature.

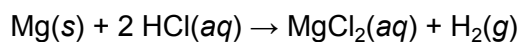
Time spent: 0 sec

Subject:
General Chemistry

QID: 403177

System:
Thermodynamics, Kinetics & Gas Laws

Magnesium metal reacts with hydrochloric acid according to Reaction 1.



Reaction 1

During an experiment to measure the molar enthalpy ΔH° of Reaction 1, a student placed 100. mL of dilute HCl(aq) solution into a coffee cup calorimeter, as shown in Figure 1.

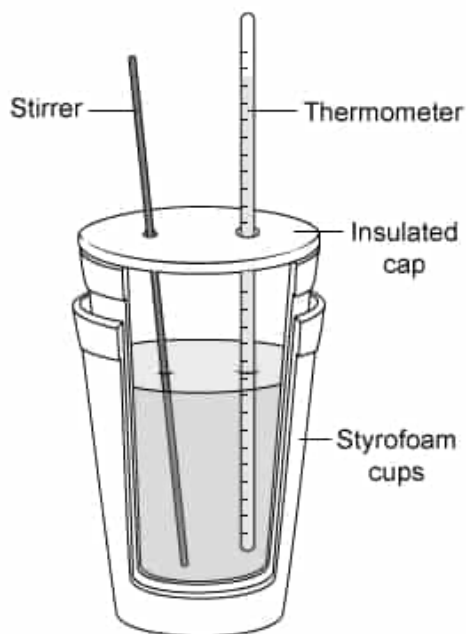


Figure 1 Coffee cup calorimeter

After monitoring the HCl(aq) solution temperature for 1 min, 0.486 g (0.0200 mol) of Mg(s) was then added to the HCl(aq) solution inside the calorimeter and stirred while continuing to monitor the temperature until the Mg(s) was completely consumed. The results of the experiment are shown on the graph in Figure 2.

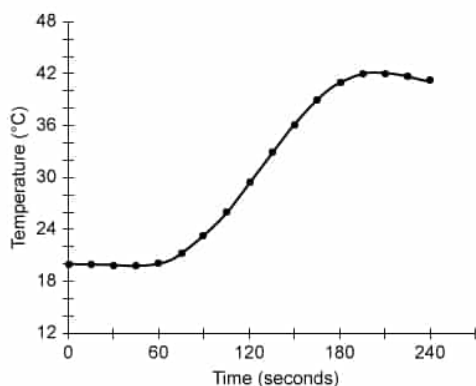


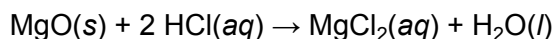
Figure 2 Graph of temperature versus time during the reaction of magnesium with hydrochloric acid

According to the first law of thermodynamics, the heat produced (or consumed) by a reaction q_{rxn} must be equal to the heat transferred to (or taken from) the surroundings (ie, the solution and the calorimeter), as expressed by Equation 1.

$$-q_{\text{rxn}} = q_{\text{surroundings}} = q_{\text{solution}} + q_{\text{calorimeter}}$$

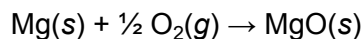
Equation 1

The dissolution of magnesium oxide in hydrochloric acid is represented by Reaction 2:

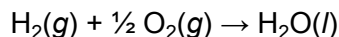


Reaction 2: $\Delta H^\circ = +X \text{ kJ/mol}$

and the formation for magnesium oxide and water are represented by Reactions 3 and 4, respectively.



Reaction 3: $\Delta H^\circ = -624 \text{ kJ/mol}$



Reaction 4: $\Delta H^\circ = -286 \text{ kJ/mol}$

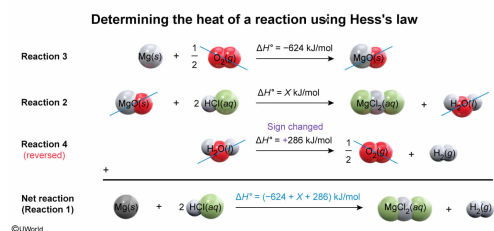
Which expression is equivalent to the molar enthalpy of Reaction 1 (ΔH°) from the calorimetry experiment in the passage?

- ☐ A. $\Delta H^\circ = (-624 - X - 286) \text{ kJ/mol}$ [5%]
☐ B. $\Delta H^\circ = (-624 - X + 286) \text{ kJ/mol}$ [15%]
☐ C. $\Delta H^\circ = (-624 + X - 286) \text{ kJ/mol}$ [23%]
✓ ☒ D. $\Delta H^\circ = (-624 + X + 286) \text{ kJ/mol}$ [55%]

Correct

55% answered correctly

Explanation:



The **change in the total energy** of a system is indicated by its **change in enthalpy** ΔH° . When pressure does not change, ΔH is equal to the amount of heat that is absorbed or released by a reaction. A positive ΔH indicates an **endothermic** reaction that **absorbs heat** whereas a negative ΔH indicates an **exothermic** reaction that **releases heat**.

Hess's law states that if a reaction can be performed in more than one step, the **heat of the overall reaction** (its overall enthalpy) is equal to the **sum of the enthalpy changes** from each step. Accordingly, Reaction 1 from the passage can be achieved by the sum of Reaction 2, Reaction 3, and the reverse of Reaction 4. In the summation, chemical species on opposite sides of the reaction arrows cancel. Because Reaction 4 is reversed, the sign of ΔH for the reaction must be switched from negative to positive.

Therefore, the expression $\Delta H^\circ = (-624 + X + 286) \text{ kJ/mol}$ is equivalent to the molar enthalpy of Reaction 1.

(Choices A and B) When adding Reaction 2, Reaction 3, and Reaction 4 to achieve a net reaction equivalent to Reaction 1, the direction of Reaction 2 should not be reversed. As such, the sign of the

enthalpy of Reaction 2 should not be switched and should remain *positive* (ie, +X kJ/mol) in the summation.

(Choice C) To achieve a net reaction equivalent to Reaction 1, the direction of Reaction 4 must be reversed. Consequently, the sign of ΔH° for Reaction 4 must be made *positive* (ie, +286 kJ/mol) in the summation.

Educational objective:

According to Hess's law, the overall change in enthalpy ΔH of a reaction is equal to the sum of the enthalpy changes from each step of the reaction. When summing reaction steps, if a reaction is reversed, its ΔH has the same value but the opposite sign, and if a reaction is scaled, both the reaction and its ΔH must be multiplied (or divided) by the same factor.

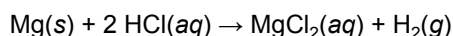
Time spent:0 sec

Subject:
General Chemistry

QID:403178

System:
Thermodynamics, Kinetics & Gas Laws

Magnesium metal reacts with hydrochloric acid according to Reaction 1.



Reaction 1

During an experiment to measure the molar enthalpy ΔH° of Reaction 1, a student placed 100. mL of dilute HCl(aq) solution into a coffee cup calorimeter, as shown in Figure 1.

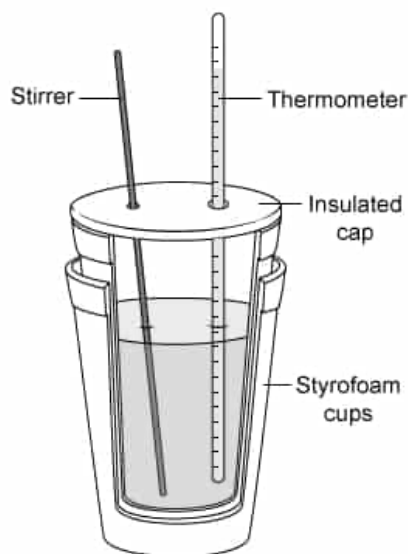


Figure 1 Coffee cup calorimeter

After monitoring the HCl(aq) solution temperature for 1 min, 0.486 g (0.0200 mol) of Mg(s) was then added to the HCl(aq) solution inside the calorimeter and stirred while continuing to monitor the temperature until the Mg(s) was completely consumed. The results of the experiment are shown on the graph in Figure 2.

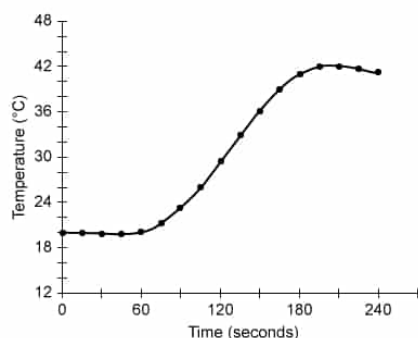


Figure 2 Graph of temperature versus time during the reaction of magnesium with hydrochloric acid

According to the first law of thermodynamics, the heat produced (or consumed) by a reaction q_{rxn} must be equal to the heat transferred to (or taken from) the surroundings (ie, the solution and the calorimeter), as expressed by Equation 1.

$$-q_{\text{rxn}} = q_{\text{surroundings}} = q_{\text{solution}} + q_{\text{calorimeter}}$$

Equation 1

What is the total volume of the hydrogen gas produced by Reaction 1 during the calorimetry experiment if the gas volume is measured at standard temperature and pressure (STP)? (Note: Assume that all the Mg metal reacts.)

- ☐ A. $8.92 \times 10^{-4} \text{ L}$ [4%]
- ☐ B. $2.24 \times 10^{-1} \text{ L}$ [31%]
- ☒ C. $4.48 \times 10^{-1} \text{ L}$ [59%]
- ☐ D. $2.65 \times 10^2 \text{ L}$ [3%]

Correct

59% answered correctly

Explanation:

Volume of gas produced from a reaction at STP

$\text{Mg(s)} + 2 \text{HCl(aq)} \rightarrow \text{MgCl}_2\text{(aq)} + \text{H}_2\text{(g)}$

Mass Mg(s) reacting	Molar mass of Mg(s)	Mole ratio of reaction	Mole volume of gas at STP	Volume of H ₂ (g) produced at STP
0.486 g Mg(s)	$\frac{1 \text{ mol Mg(s)}}{24.31 \text{ g Mg(s)}}$	$\frac{1 \text{ mol H}_2\text{(g)}}{1 \text{ mol Mg(s)}}$	$\frac{22.4 \text{ L H}_2\text{(g)}}{1 \text{ mol H}_2\text{(g)}}$	0.448 L H ₂ (g)

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Dividing the given mass of Mg(s) by its **molar mass** gives the moles of Mg(s) involved in Reaction 1:

$$0.486 \text{ g Mg(s)} \times \frac{1 \text{ mol}}{24.31 \text{ g Mg(s)}} = 0.0200 \text{ mol Mg(s)}$$

In a reaction, the moles of one chemical species can be related to the moles of other chemical species by applying **mole ratios** obtained from the **stoichiometric coefficients** of a **balanced reaction equation**. Reaction 1 shows that the number of moles of H₂(g) produced is equal to the number of moles of Mg(s) that react:

$$0.0200 \text{ mol Mg(s)} \times \frac{1 \text{ mol H}_2\text{(g)}}{1 \text{ mol Mg(s)}} = 0.0200 \text{ mol H}_2\text{(g)}$$

Under conditions of **standard temperature and pressure (STP)**, which is defined as **273 K** (0 °C) and **1 atm** of pressure, a gas occupies a volume of **22.4 L for every 1 mole** of gas molecules. (This molar volume of 22.4 L/mol can be confirmed by substituting the STP values into the **ideal gas law** and solving for V/n.) Accordingly, calculations for a gas at STP can be made by using this molar volume ratio as a conversion factor:

$$0.0200 \text{ mol H}_2\text{(g)} \times \frac{22.4 \text{ L H}_2\text{(g)}}{1 \text{ mol H}_2\text{(g)}} = 0.448 \text{ L H}_2\text{(g)}$$

Therefore, the total volume of H₂(g) produced by Reaction 1 during the calorimetry experiment (measured at STP) is 4.48 × 10⁻¹ L.

(Choice A) A value of 8.92 × 10⁻⁴ (with incorrect units) results from dividing instead of multiplying the moles of H₂(g) by 22.4 L/mol in the last step of the calculation.

(Choice B) 2.24 × 10⁻¹ L is the volume occupied by 0.0100 moles of H₂(g) at STP. This value results from using the wrong mole ratio in the second step of the calculation.

(Choice D) A value of 2.65 × 10² (with incorrect units) results from multiplying instead of dividing the mass of Mg(s) by its molar mass in the first step of the calculation.

Educational objective:

Under conditions of standard temperature and pressure (STP), which are defined as 273 K (0 °C) and 1 atm of pressure, 1 mole of a gas occupies a volume of 22.4 L. Calculations for a gas at STP can be made by using this molar volume ratio as a conversion factor.

Time spent: 0 sec

Subject:
General Chemistry

QID: 403179

System:
Thermodynamics, Kinetics & Gas Laws